

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 June 2023

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2023 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. In humans, fertilisation normally occurs in the **[1]**

- A) uterus wall only
- B) ovary
- C) cervix
- D) oviduct / Fallopian tube

2. The SI unit of speed is **[1]**

- A) m/s
- B) J
- C) N
- D) kg

3. Liquid pressure increases with **[1]**

- A) only the mass of the observer
- B) magnetic field only
- C) colour of the container
- D) depth and density of the liquid

4. Electrolysis requires **[1]**

- A) a solid ionic compound with no mobile ions
- B) pure water with no ions, always
- C) a molten or aqueous ionic compound
- D) a sugar solution

5. Pressure is defined as **[1]**

- A) $F + A$
- B) $F \times A$
- C) A / F
- D) F / A

6. A magnetic material can be magnetised by **[1]**

- A) heating it in water only
- B) stroking with a magnet or placing it in a coil with d.c.
- C) painting it red
- D) dissolving it

7. Chlorine is added to drinking water to [1]

- A) kill microorganisms
- B) make it cloudy
- C) remove all ions
- D) add hardness

8. A herbivore is a [1]

- A) pathogen
- B) primary consumer
- C) producer
- D) decomposer only

9. A dull black surface, compared with a shiny silver surface, [1]

- A) reflects all heat always
- B) is a poor absorber of radiation
- C) is a good absorber and emitter of radiation
- D) cannot emit radiation

10. The large permanent vacuole in a plant cell [1]

- A) makes cellulose
- B) controls cell division
- C) contains cell sap and helps keep the cell turgid
- D) is the site of respiration

11. When light travels from air into glass it [1]

- A) slows down and bends towards the normal
- B) does not change direction ever
- C) speeds up and bends away from the normal
- D) stops

12. A gas has [1]

- A) particles that cannot move
- B) fixed shape and volume
- C) no fixed shape or volume
- D) fixed volume, no fixed shape

13. The period number of an element is the number of [1]

- A) neutrons
- B) outer electrons
- C) occupied electron shells
- D) protons only

14. Water enters a root hair cell mainly by [1]

- A) photosynthesis
- B) transpiration pull only
- C) ejection
- D) osmosis

15. Metals are good thermal conductors because they have [1]

- A) only covalent bonds
- B) a vacuum inside
- C) free electrons that transfer energy
- D) no particles

16. The main function of red blood cells is to **[1]**

- A) fight all pathogens by phagocytosis only
- B) clot blood
- C) transport oxygen
- D) make urea

17. Distilled water, compared with tap water, contains **[1]**

- A) fewer dissolved salts
- B) sand
- C) more dissolved salts
- D) more chlorine always

18. Tuberculosis (TB) is caused by a bacterium and is spread mainly by **[1]**

- A) mosquitoes
- B) ticks
- C) contaminated rivers only
- D) droplet infection (coughing, sneezing)

19. In a solid, particles are typically **[1]**

- A) far apart and move freely
- B) close together and vibrate about fixed positions
- C) close and slide over each other
- D) not attracted at all

20. Phloem tissue transports **[1]**

- A) soil particles
- B) water only upwards
- C) sucrose and amino acids
- D) oxygen

21. A pollutant gas from burning coal that causes acid rain is **[1]**

- A) nitrogen (as N₂)
- B) argon
- C) sulfur dioxide
- D) oxygen

22. Sound cannot travel through **[1]**

- A) air
- B) a vacuum
- C) water
- D) steel

23. Hydrochloric acid in the stomach **[1]**

- A) kills many bacteria and provides a low pH for protease
- B) digests cellulose
- C) stores bile
- D) neutralises bile

24. Which gas is produced during photosynthesis? **[1]**

- A) hydrogen
- B) oxygen
- C) methane
- D) nitrogen

25. The most abundant gas in clean dry air is **[1]**

- A) carbon dioxide
- B) nitrogen
- C) oxygen
- D) argon

26. A mass of 5 kg accelerates at 3 m/s^2 . The unbalanced force is **[1]**

- A) 0 N
- B) 50 N
- C) 15 N
- D) 8 N

27. The rate of a reaction increases when **[1]**

- A) temperature decreases
- B) the surface area of a solid reactant increases
- C) the catalyst is removed
- D) concentration decreases

28. Average speed is **[1]**

- A) mass / time
- B) distance / time
- C) time / distance
- D) distance $\times$ time

29. The period number of an element is the number of **[1]**

- A) protons only
- B) outer electrons
- C) neutrons
- D) occupied electron shells

30. An object has mass 20 g and volume 10 cm^3 . Its density is **[1]**

- A) 30 g/cm^3
- B) 10 g/cm^3
- C) 200 g/cm^3
- D) 2 g/cm^3

31. In human muscle during vigorous exercise, anaerobic respiration produces **[1]**

- A) lactic acid
- B) ethanol only
- C) oxygen
- D) carbon dioxide and water

32. The anode is the **[1]**

- A) negative electrode
- B) electrolyte
- C) positive electrode
- D) ammeter

33. Which energy conversion is correct? **[1]**

- A) gravitational potential $\rightarrow$ kinetic as an object falls
- B) chemical $\rightarrow$ kinetic in a battery at rest
- C) light $\rightarrow$ mass always
- D) sound $\rightarrow$ nuclear always

34. White blood cells [1]

- A) transport sucrose
- B) are fragments that clot blood
- C) are involved in defence against pathogens
- D) carry oxygen only

35. Which statement is correct? [1]

- A) Arteries carry blood away from the heart
- B) Capillaries have thick muscular walls
- C) Veins have the highest blood pressure
- D) Xylem carries blood

36. Iron rusts when it is in contact with [1]

- A) nitrogen only
- B) oxygen and water
- C) argon
- D) oil only

37. After fertilisation in a flower, the ovule becomes the [1]

- A) petal
- B) anther
- C) sepal
- D) seed

38. The SI unit of mass is the [1]

- A) kilogram
- B) degree Celsius
- C) metre per second
- D) newton

39. A lighted splint is used to test for hydrogen. A positive result is [1]

- A) a glowing splint relighting
- B) limewater turning milky
- C) bleaching litmus
- D) a pop sound

40. Red blood cells are adapted to carry oxygen because they contain [1]

- A) cellulose
- B) starch
- C) haemoglobin
- D) chlorophyll

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).